

An Approximate Capacity Bound for Sparse Combinatorial Key-Value Memory: Full Derivation and Validation

Supplementary note to “Combinatorial Coding, Key-Value Recall, and Emergent
Constraint Satisfaction in a Hippocampal-Inspired Spiking Network”

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Abstract

This note gives the complete derivation, with every intermediate step made explicit, of the approximate capacity bound used in the main paper (Section 2.3): an expression for the largest number of items M a sparse K -of- H combinatorial key-value memory can store while remaining reliably retrievable under a given level of query corruption. The bound is structurally the same as modern Hopfield networks’ proven exponential capacity [1], but is derived here for a discrete, sparse code via elementary combinatorics and extreme value theory rather than for dense continuous patterns via an energy-function argument. We include a full, self-contained proof of the one non-elementary step (the expected maximum of many correlated-in-effect-but-treated-as-independent Gaussian similarities), validate the final bound against simulation, and discuss its relationship to the wider family of capacity theorems it belongs to. We then go further than that first-moment approximation: Section 8 derives an exact, non-asymptotic failure probability directly from the hypergeometric distribution, and a corresponding rigorous sufficient bound on capacity at any chosen target reliability δ , correcting a subtle correlation error that a natural-looking marginalization over query corruption would otherwise introduce. This exact bound is, as expected, considerably more conservative than the first-moment approximation, and is shown to be tight in the well-separated regime and honestly, quantifiably loose only as the system approaches its collapse transition.

1 Setup

We store M items. Item m ’s key is a binary vector $h_m \in \{0, 1\}^H$ with exactly K active (value-1) coordinates – a K -of- H sparse code. Throughout this note we model the M stored keys as independent, uniformly random K -subsets of $[H] = \{1, \dots, H\}$: an idealization relative to the true architecture, where keys are W_{SH} -projections of mutually correlated combinatorial structure (solved Sudoku grids), addressed explicitly in Section 9.

A query is formed from the true key h_{true} by flipping a fraction $\eta \in [0, 1]$ of all H coordinates, chosen uniformly at random without replacement – the noise model used throughout the main paper. Retrieval compares the query against all M stored keys via similarity (shared active coordinates) and should return the index of h_{true} . Retrieval *fails* when some other stored key’s similarity to the query exceeds h_{true} ’s.

We ask: for fixed H , K , and corruption level η , how large can M be before this kind of failure becomes likely?

2 Step 1: Similarity between two independent random keys

Lemma 1 (Hypergeometric overlap). *Let $A, B \subset [H]$ be independent uniformly random subsets with $|A| = a$, $|B| = b$. Then $|A \cap B|$ is a hypergeometric random variable with*

$$\mathbb{E}[|A \cap B|] = \frac{ab}{H}, \quad \text{Var}(|A \cap B|) = a \cdot \frac{b}{H} \left(1 - \frac{b}{H}\right) \cdot \frac{H - a}{H - 1}.$$

Proof sketch. Fix A and treat B 's b elements as drawn sequentially without replacement from $[H]$. Let $X_i = \mathbb{1}[\text{the } i\text{-th draw lands in } A]$, so $|A \cap B| = \sum_{i=1}^b X_i$. Each X_i is Bernoulli(a/H) by symmetry, giving the mean by linearity of expectation. For the variance, $\text{Cov}(X_i, X_j) < 0$ for $i \neq j$ because sampling without replacement makes draws negatively associated (having drawn a marked element makes drawing another marked element on a later draw slightly less likely); summing the b variances and $\binom{b}{2}$ covariances gives the stated finite-population-corrected formula. (Full derivation: Feller, *An Introduction to Probability Theory*, Vol. 1, Ch. II.) $\square$

Applying Lemma 1 with $a = b = K$ gives the similarity between two *uncorrupted* stored keys: mean K^2/H , the number quoted throughout the main paper (e.g. $K^2/H = 22.5$ for the default $H = 1000$, $K = 150$).

3 Step 2: Statistics of a corrupted query

Flipping a uniformly random fraction η of all H coordinates is itself an instance of Lemma 1: the ηH flipped positions are a random subset of $[H]$, and we ask how many of h_{true} 's K active positions they contain. By the lemma (with $a = K$, $b = \eta H$), the expected number of active positions flipped *off* is ηK , so

$$S_{\text{true}}(\eta) = \mathbb{E}[\text{similarity of query to } h_{\text{true}}] = K(1 - \eta), \quad (1)$$

exactly (no further approximation needed: this is a direct expectation, not a bound). The expected number of originally-*inactive* positions flipped *on* is, symmetrically, $\eta(H - K)$, so the corrupted query's own total active-coordinate count is, in expectation,

$$K_q(\eta) = K(1 - \eta) + \eta(H - K) = K(1 - 2\eta) + \eta H. \quad (2)$$

The query's similarity to any *other*, independent stored key is now Lemma 1 applied with $a = K_q(\eta)$, $b = K$:

$$\mu_{\text{other}}(\eta) = \frac{K_q(\eta) K}{H}, \quad \sigma_{\text{other}}^2(\eta) = K_q(\eta) \frac{K}{H} \left(1 - \frac{K}{H}\right) \frac{H - K_q(\eta)}{H - 1}. \quad (3)$$

4 Step 3: Gaussian approximation

The hypergeometric distribution of S_{other} (similarity to one specific other key) is exact but discrete and unwieldy for the extreme-value step that follows. We approximate

$$S_{\text{other}} \approx \mathcal{N}(\mu_{\text{other}}(\eta), \sigma_{\text{other}}^2(\eta)),$$

justified for $H \gg 1$ by the central limit theorem for sampling without replacement from a finite population [2, 3]: sums arising from simple random sampling without replacement are asymptotically normal as the population size grows, with exactly the finite-population-corrected variance already appearing in Lemma 1. This is the least exact step in the derivation, and its error is largest precisely when σ_{other} is small relative to the mean – i.e. at low noise, where the discreteness of the underlying hypergeometric distribution matters most. Section 7 quantifies this error directly rather than leaving it as a caveat.

5 Step 4: The maximum of many competitors

Retrieval does not fail when the *typical* competing key looks similar to the query – it fails when the *single most similar* of the $M - 1$ competitors exceeds $S_{\text{true}}(\eta)$. This is an extreme-value criterion, and it is the step that most distinguishes this derivation from a naive signal-to-noise argument.

Lemma 2 (Expected maximum of i.i.d. Gaussians). *Let $Z_1, \dots, Z_n$ be i.i.d. $\mathcal{N}(0, 1)$. Then*

$$\mathbb{E} \left[\max_{1 \leq i \leq n} Z_i \right] \leq \sqrt{2 \ln n} + o(1) \quad \text{as } n \rightarrow \infty,$$

and this bound is asymptotically tight: $\mathbb{E}[\max_i Z_i] = \sqrt{2 \ln n} (1 + o(1))$.

Proof of the upper bound. For any $t \geq 0$, the standard Gaussian tail (Chernoff) bound gives $\Pr(Z_1 > t) \leq e^{-t^2/2}$: with $s = t$, $\Pr(Z_1 > t) = \Pr(e^{sZ_1} > e^{st}) \leq \mathbb{E}[e^{sZ_1}]e^{-st} = e^{s^2/2-st}$, minimized at $s = t$. A union bound over the n variables gives $\Pr(\max_i Z_i > t) \leq \min(1, ne^{-t^2/2})$. Writing the expectation of a nonnegative random variable as an integral of its tail,

$$\mathbb{E} \left[\max_i Z_i \right] = \int_0^\infty \Pr \left(\max_i Z_i > t \right) dt \leq \int_0^{t^*} 1 dt + \int_{t^*}^\infty ne^{-t^2/2} dt,$$

where $t^* = \sqrt{2 \ln n}$ is chosen so the two bounds on the integrand meet ($ne^{-(t^*)^2/2} = 1$). The first integral is exactly $t^* = \sqrt{2 \ln n}$. For the second, the standard Gaussian tail estimate $\int_a^\infty e^{-t^2/2} dt \leq \frac{1}{a} e^{-a^2/2}$ (for $a > 0$, obtained by writing $e^{-t^2/2} \leq \frac{t}{a} e^{-t^2/2}$ on $t \geq a$ and integrating exactly) gives

$$\int_{t^*}^\infty ne^{-t^2/2} dt \leq \frac{n}{t^*} e^{-(t^*)^2/2} = \frac{1}{\sqrt{2 \ln n}} \xrightarrow{n \rightarrow \infty} 0.$$

Summing the two pieces gives $\mathbb{E}[\max_i Z_i] \leq \sqrt{2 \ln n} + o(1)$. The matching lower bound requires a second-moment (Paley–Zygmund-type) argument and is omitted here; see Boucheron, Lugosi & Massart, *Concentration Inequalities* [4], Ch. 2, for the complete two-sided result. $\square$

Rescaling Lemma 2 to $\mathcal{N}(\mu_{\text{other}}, \sigma_{\text{other}}^2)$ variables and setting $n = M - 1$ (the number of competing stored keys) gives the working approximation used in the main paper:

$$\mathbb{E} \left[\max_{i=1}^{M-1} S_{\text{other},i} \right] \approx \mu_{\text{other}}(\eta) + \sigma_{\text{other}}(\eta) \sqrt{2 \ln(M-1)}. \quad (4)$$

6 Step 5: Solving for capacity

Retrieval is (approximately, at the level of expectations rather than a proper tail bound on failure probability) reliable when $S_{\text{true}}(\eta)$ exceeds the expected maximum competitor, Equation 4. Define the normalized separation margin

$$z(\eta) = \frac{S_{\text{true}}(\eta) - \mu_{\text{other}}(\eta)}{\sigma_{\text{other}}(\eta)}. \quad (5)$$

Setting $S_{\text{true}}(\eta) = \mu_{\text{other}}(\eta) + \sigma_{\text{other}}(\eta) \sqrt{2 \ln(M-1)}$ and solving for M :

$$z(\eta) = \sqrt{2 \ln(M-1)} \iff M-1 = \exp\left(\frac{z(\eta)^2}{2}\right) \iff \boxed{M_{\text{max}}(\eta) \approx 1 + \exp\left(\frac{z(\eta)^2}{2}\right)}. \quad (6)$$

Proposition 1 (Structural form). *Equation 6 states that capacity is exponential in the square of a normalized signal-to-noise-like quantity. This is the same structural form as modern Hopfield networks’ proven capacity $C \cong 2^{d/2}$ [1], though that bound is derived via a Lyapunov energy-function argument (the Concave–Convex Procedure applied to a log-sum-exp energy) over dense continuous patterns, not via hypergeometric statistics over a sparse discrete code.*

7 Validation against simulation

Equations 1–6 were implemented exactly as derived (`capacity_bound_derivation.py`, no free parameters fit to data) and compared against the empirically measured collapse thresholds reported in the main paper.

	$M=50$	$M=1000$	$M=10,000$	$K=10$	$K=150$
Predicted (this note, $H=1000$)	43.9%	41.9%	40.6%	24.5%	41.9%
Observed (simulation)	45%	45%	40%	25%	40–45%

Agreement is within 1–5 percentage points across a 200-fold range of M and an order of magnitude in K , including an almost exact match at $K = 10$. A second, independent check: at $H=1000$, $K=150$, $M=200$, the derivation predicts the ratio $S_{\text{true}}(\eta)/\mathbb{E}[\max_i S_{\text{other},i}]$ at $\eta \in \{0, 0.20, 0.30\}$ as 4.21, 1.99, 1.47 respectively, against independently measured ratios of 3.39, 1.92, 1.45 (obtained from a completely separate experiment, the separation-operator ablation, not used to fit any parameter of this derivation). The two noisy points match to within 4% and 1.4%; the noise-free point is over-predicted by about 24%, consistent with Section 4’s observation that the Gaussian approximation is weakest exactly there.

8 An exact hypergeometric tail bound

Equation 6 matches only the *expected* maximum competitor similarity, not a probabilistic guarantee on retrieval failure – a gap noted below in Section 9. This section closes it with an exact, non-asymptotic tail bound on the failure probability itself, derived directly from the hypergeometric distribution rather than its Gaussian approximation, and valid at any finite M .

8.1 A correlation a naive marginalization misses

Let R denote the (random) number of h_{true} ’s K active positions that land among the ηH flipped query positions. By Lemma 1 (with $a = K$, $b = \eta H$), R is exactly hypergeometric:

$$R \sim \text{Hypergeometric}(H, K, \eta H), \quad \Pr(R = r) = \frac{\binom{K}{r} \binom{H-K}{\eta H - r}}{\binom{H}{\eta H}}.$$

Conditional on $R = r$, both the query’s active-coordinate count and its similarity to h_{true} are *deterministic*, not merely expected:

$$k(r) = K + \eta H - 2r, \quad s(r) = K - r,$$

the realized versions of Equations 2 and 1, which recover $k(\eta K) = K_q(\eta)$ and $s(\eta K) = S_{\text{true}}(\eta)$ only in expectation. Crucially, $k(r)$ and $s(r)$ are perfectly correlated through the same draw of r :

a query realization with more of its true bits flipped off (large r) simultaneously has lower true-similarity *and* a different active count. A natural-looking but invalid shortcut is to hold $s(r)$ fixed at its unconditional expectation $S_{\text{true}}(\eta) = K(1 - \eta)$ while still letting $k(r)$ vary with r inside the union bound over competitors: since $k(r)$ and $s(r)$ are deterministic functions of the very same random variable, fixing one while randomizing the other breaks their correlation and is not a valid marginalization of any well-defined random process. Numerically, this shortcut understates the true failure probability by roughly four to five orders of magnitude for $\eta \in [0.30, 0.40]$, converging to the correct value only as η approaches the collapse region (where r 's distribution concentrates and the distinction between fixing $s(r)$ at r or at $\mathbb{E}[r]$ stops mattering).

8.2 Exact per-realization failure probability

Fix one specific other stored key h_j ($j \neq \text{true}$), itself an independent random K -subset of $[H]$. Conditional on $R = r$, the query is a specific, realized $k(r)$ -subset, and its overlap with h_j is exactly hypergeometric by Lemma 1 (with $a = k(r)$, $b = K$). Define the exact tail probability

$$p_k(s) := \Pr[\text{Hypergeometric}(H, k, K) \geq s] = \sum_{i=s}^{\min(k, K)} \frac{\binom{k}{i} \binom{H-k}{K-i}}{\binom{H}{K}}.$$

(We use $\geq$ rather than strict $>$: an exact tie is conservatively counted as a failure, which can only inflate the resulting probability and keeps Proposition 2 below a genuine upper bound regardless of how ties are broken at retrieval time.) Conditional on $R = r$, the probability that this one specific competitor's similarity meets or exceeds h_{true} 's is exactly $p_{k(r)}(s(r))$, and with $M - 1$ independent competitors the conditional failure probability is exactly

$$\Pr[\text{fail} \mid R = r] = 1 - [1 - p_{k(r)}(s(r))]^{M-1}.$$

Averaging over R gives the exact unconditional failure probability – no Gaussian approximation, no expectation-of-a-maximum heuristic, and no fixing of $s(r)$ at its mean:

$$P_{\text{fail}}(M, \eta) = \sum_{r=0}^K \Pr(R = r) \cdot \left\{ 1 - [1 - p_{k(r)}(s(r))]^{M-1} \right\}. \quad (7)$$

Equation 7 is exact given the independence assumption of Section 9; it involves only a finite sum over $r \in \{0, \dots, K\}$ and is cheap to evaluate for any H, K, η, M .

8.3 A rigorous, closed-form sufficient bound

Equation 7 is exact but implicit: it gives P_{fail} as a function of M , not M as a function of a target failure probability δ . We invert it via the union bound, applied *per realization of r* before averaging – the order matters, and doing it in this order is what keeps the resulting bound valid.

By Bernoulli's inequality, $1 - (1 - p)^n \leq np$ for any $p \in [0, 1]$ and integer $n \geq 0$. Applied pointwise, for every r ,

$$1 - [1 - p_{k(r)}(s(r))]^{M-1} \leq (M - 1) p_{k(r)}(s(r)).$$

This is a deterministic inequality between two functions of r ; taking $\mathbb{E}_R[\cdot]$ of both sides preserves it, since expectation is monotone:

$$P_{\text{fail}}(M, \eta) \leq (M - 1) \cdot \underbrace{\mathbb{E}_R[p_{k(R)}(s(R))]}_{=: \bar{P}(\eta)}. \quad (8)$$

Proposition 2 (Corrected sufficient capacity bound). *For any target failure probability $\delta \in (0, 1)$, storing*

$$M \leq M_{\max}(\eta, \delta) := 1 + \left\lfloor \frac{\delta}{\bar{P}(\eta)} \right\rfloor \quad (9)$$

items guarantees $P_{\text{fail}}(M, \eta) \leq \delta$ exactly, subject only to the independence assumption of Section 9 – not to any Gaussian or large- M asymptotic approximation.

Proof. Immediate from Equation 8: $(M - 1)\bar{P}(\eta) \leq \delta$ whenever $M - 1 \leq \delta/\bar{P}(\eta)$, i.e. $M \leq 1 + \lfloor \delta/\bar{P}(\eta) \rfloor$. $\square$

Equation 9 has the same union-bound shape as a natural first attempt at this formula, $1 + \lfloor \delta/p_{K_q(\eta)}(S_{\text{true}}(\eta)) \rfloor$, but with p correctly taken to be $\bar{P}(\eta) = \mathbb{E}_R[p_{k(R)}(s(R))]$ – the per-realization tail probability *properly averaged* over R – rather than a single tail probability evaluated at the mismatched pair $(K_q(\eta), S_{\text{true}}(\eta))$, which (Section 8.1) treats the numerator and denominator of the same random draw as independently settable.

8.4 Numerical validation

Table 1 reports $\bar{P}(\eta)$ and the resulting $M_{\max}(\eta, \delta)$ at $H = 1000$, $K = 150$ for two target failure probabilities, alongside $M_{\max}(\eta)$ from Equation 6 (Section 6) for comparison. All quantities were computed with `scipy.stats.hypergeom` and, where probabilities are extremely small, the numerically stable form $1 - (1 - p)^n = -\text{expm1}(n \cdot \log1p(-p))$ to avoid catastrophic cancellation.

η	$\bar{P}(\eta)$	$M_{\max}(\eta, \delta=0.01)$	$M_{\max}(\eta, \delta=0.05)$	$M_{\max}(\eta)$ [Eq. 6]
0.20	2.41×10^{-35}	4.15×10^{32}	2.07×10^{33}	2.29×10^{48}
0.30	8.26×10^{-15}	1.21×10^{12}	6.05×10^{12}	1.61×10^{19}
0.35	9.59×10^{-9}	1.04×10^6	5.21×10^6	2.61×10^{10}
0.40	1.22×10^{-4}	83.0	411	3.27×10^4
0.42	1.81×10^{-3}	6.0	28.0	739
0.45	3.73×10^{-2}	1.0	2.0	13.9

Table 1: Corrected exact sufficient bound versus the expected-maximum approximation (both at $H = 1000$, $K = 150$).

Two things stand out. First, the exact tail-based bound is dramatically more conservative than the expected-maximum approximation – by about seven orders of magnitude at $\eta = 0.30$ (1.2×10^{12} vs. 1.6×10^{19}), narrowing to about a factor of 120 at $\eta = 0.42$. This gap is expected, not a contradiction between the two derivations: Equation 6 only requires the true match to beat the *expected value* of the maximum competitor, a much weaker requirement than controlling the actual *probability* of failure to a small δ via a union bound over $M - 1$ competitors. The two bounds answer different questions – “typical-case margin” versus “guaranteed failure rate” – and are best read together: Equation 6 explains *why* capacity is exponential in the squared margin (the mechanism), while Equation 9 gives an actually-defensible number at a chosen reliability level (the guarantee). Section 9’s prediction that a proper tail bound “would very likely give a somewhat smaller $M_{\max}$ ” is confirmed, considerably more starkly than that hedge anticipated.

Second, we check how loose the sufficient bound (Equation 9) is relative to the exact failure probability it bounds (Equation 7), by comparing $(M - 1)\bar{P}(\eta)$ against $P_{\text{fail}}(M, \eta)$ directly (Table 2).

The bound is essentially tight (ratio ≈ 1.00) at $\eta = 0.30$, and increasingly loose – by design, as any union bound must be once per-competitor probabilities are no longer small – as η approaches

η	M	$P_{\text{fail}}(M, \eta)$ [exact, Eq. 7]	$(M - 1)\bar{P}(\eta)$ [Eq. 8]	ratio
0.30	200	1.644×10^{-12}	1.644×10^{-12}	1.00
0.30	1000	8.252×10^{-12}	8.252×10^{-12}	1.00
0.30	4000	3.303×10^{-11}	3.303×10^{-11}	1.00
0.40	200	1.82×10^{-2}	2.42×10^{-2}	1.33
0.40	1000	5.98×10^{-2}	1.22×10^{-1}	2.03
0.40	4000	1.35×10^{-1}	4.87×10^{-1}	3.59
0.42	200	1.59×10^{-1}	3.61×10^{-1}	2.27
0.42	1000	3.46×10^{-1}	1.81×10^0	5.23
0.42	4000	5.38×10^{-1}	7.25×10^0	13.46

Table 2: Tightness of the sufficient bound against the exact failure probability.

the collapse transition and M grows (up to a factor of 13.5 at $\eta = 0.42$, $M = 4000$). This is the expected and honest behavior of a union bound: tightest exactly in the well-separated regime where a capacity claim is most meaningful, and openly announcing its own growing conservatism as the system approaches saturation rather than failing silently.

9 Limitations

Two assumptions are false by construction and are stated as such rather than absorbed silently.

Independence. Real stored keys are W_{SH} -projections of genuinely correlated combinatorial objects (solved Sudoku grids share row/column/box structure), not independent uniform random subsets. The observed collapse thresholds sit consistently at or slightly above the predicted ones (Section 7); a plausible explanation is that this real correlation makes the worst-case competing key marginally less extreme than the independent model assumes, though this has not been tested directly.

The extreme-value step is a first-moment approximation, not a tail bound – now resolved. Equation 4 matches the *expected* maximum, not a high-probability bound on it. Section 8 closes this gap with an exact, non-asymptotic alternative (Equation 7) and a corresponding rigorous sufficient bound (Proposition 2), derived directly from the hypergeometric distribution rather than the Gaussian approximation or the expected-maximum heuristic. As anticipated, the rigorous bound gives a substantially smaller M_{max} at any fixed target failure probability – by several orders of magnitude away from the collapse region (Table 1) – and is itself provably tight in that same well-separated regime, growing conservatively (rather than silently failing) only as η approaches collapse (Table 2). Equation 6 should now be read as characterizing the right *mechanism* – exponential capacity in the squared separation margin, hence a sharp rather than gradual collapse – while Equation 9 is the bound to use for an actual reliability guarantee at a chosen δ . Both bounds still share the independence assumption below, which remains open.

10 Relation to the wider literature

This derivation belongs to a family of results sharing one underlying question: does a correct answer stand out against the *extreme* value among exponentially or combinatorially many near-random alternatives? The same skeleton, worked out through different technical routes, underlies Shannon’s channel coding theorem [5], the Random Energy Model in spin-glass theory [6], Gardner’s

perceptron capacity [7], and the classical Amit–Gutfreund–Sompolinsky analysis of Hopfield network capacity [8]. Kanerva’s original capacity analysis for sparse distributed memory [9] is the closest existing derivation in spirit to this one – also a sparse binary code, also analyzed via combinatorial (there, Hamming-ball volume) arguments – though via a different geometric route than the hypergeometric-overlap statistics used here.

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